Nate B. Wiecha

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Personal Website

Research Interests: Gaussian processes, nonparametric regression, Bayesian statistics, spatial statistics, applied statistics, environmental health.

Employment

Postdoctoral Research Trainee, University of North Carolina at Chapel Hill. *2025-present*
Department of Biostatistics, Li Lab.

Research Assistant, North Carolina State University. *2022-2025*
Department of Biological Sciences, GenX Exposure Study.
Department of Statistics.

Teaching Assistant, North Carolina State University. *2021*
Department of Statistics.

Economist, Research Triangle Institute. *2016-2021*

Education

PhD, Statistics. *2025*
North Carolina State University, Raleigh, NC, USA. Advisor: Prof. Brian J. Reich.

MA, Statistics. *2021*
North Carolina State University, Raleigh, NC, USA.

BA, Economics and Political Science. *2014*
University of North Carolina at Chapel Hill, Chapel Hill, NC, USA.

Publications

N. Wiecha, E. Griffith, B.J. Reich, J. A. Hoppin (2026). When are novel methods for analyzing chemical mixtures in epidemiology beneficial? **Environmental Epidemiology**, 10(1), e456.

N. Wiecha, J. A. Hoppin, and B. J. Reich (2025). Two-stage estimators for spatial confounding with point-referenced data. **Biometrics**, 81(3), ujaf093.

N. Kotlarz, J. McCord, **N. Wiecha**, R.A. Weed, M. Cuffney, J.R. Enders, M. Strynar, D.R.U. Knappe, B.J. Reich, and J.A. Hoppin (2024). Reanalysis of PFO5DoA Levels in Blood from Wilmington, North Carolina, Residents, 2017–2018. **Environmental Health Perspectives**, 132(2), 027701.

N. Kotlarz, J. McCord, **N. Wiecha**, R.A. Weed, M. Cuffney, J.R. Enders, M. Strynar, D.R.U. Knappe, B.J. Reich, and J.A. Hoppin (2024). Measurement of Hydro-EVE and 6:2 FTS in Blood from Wilmington, North Carolina, Residents, 2017–2018. **Environmental Health Perspectives**, 132(2), 027702.

S. M. Reddy, **N. Wiecha**, C.T. Nguyen, and D.H. Barch (2023). The Role of Adverse Pregnancy Outcomes in Conventional Cardiovascular Risk Prediction. **Maternal and Child Health Journal**, 27(10), 1774-1786.

C.L. Hersey, **N. Wiecha**, L. M. Greenwald, S. M. Kissam (2022). Medicaid ACOs and Managed Care: A Tale of 2 States. **Population Health, Equity & Outcomes**, 10(3), 28-33.

Submitted manuscripts

N. Wiecha, C.E. Harrington, A. Qaqish, L.A. Eaves, M. Serre, R. Fry, H. Zhou, D. Li. Hypothesis testing for nonlinear and interactive effects using additive Gaussian processes (2026+). Submitted to **Biometrics**.

N. Wiecha, D. Batish, J.R. Enders, R.A. Weed, D. Collier, C.S. Lea, B.J. Reich, N. Kotlarz, J.A. Hoppin. Perfluoroalkyl ether alkyl acids (PFEAs), legacy PFAS, and thyroid hormone levels in PFAS-exposed North Carolina residents (2026+). Under review at **Environmental Health Perspectives**.

In preparation

N. Wiecha, D. Li (2026+). When are black-box models interpretable? **In preparation**.

N. Wiecha, A. Sauer, B.J. Reich (2026+). Gradient-based hypothesis tests in Bayesian nonparametric regression. **In preparation**.

N. Wiecha, R. Rotem, M. Weisskopf. Impacts of predictor correlation on point estimate stability and sign reversal (2026+). **In preparation**.

C. McNeill, **N. Wiecha**, Y. Baek, S. Berchuck, B.J. Reich. Addressing Spatial Confounding for Discrete Outcome Data and Areal Data (2026+). **In preparation**.

Funding

T32 Training Grant, Biostatistics for Research in Environmental Health *2025-present*
Provost Doctoral Fellowship, North Carolina State University Graduate School *2024-2025*

Presentations

Speaker, Session: "Addressing Confounding in Spatial Causal Inference." *Joint Statistical Meetings*, August 2026

Speaker, Session: "Consideration of Spatial and Temporal Variations." *Joint Statistical Meetings*, August 2025

Speaker, Session: "Advancements in Spatio-temporal Analysis and Modeling Techniques." *Eastern North American Region (ENAR) Spring Meeting*, March 2025

Speaker, Session: "Recent Applications of Gaussian Processes." *Advancements in Interdisciplinary Statistics and Combinatorics*, October 2024

Contributed Poster: "Two-stage Estimators for Spatial Confounding." *Joint Statistical Meetings*, August 2024

Speaker: "Two-stage Estimators for Spatial Confounding." *North Carolina State University Department of Statistics Seminar Series*, April 2024

Contributed Poster: "Evaluation of methods for analyzing nonlinear and counteracting associations of PFAS with health outcomes." *Superfund Research Program Annual Meeting*, December 2023

Teaching

Instructor, *Principles of Statistical Inference (BIOS 600)* *Spring 2026*
Guest Lecturer, *Epidemiology and Statistics in Global Public Health (BIO 404)* *Fall 2024, 2026*
Instructor, *Statistics PhD Qualifying Exam Prep Course* *Summer 2023*
Instructor, *Introduction to Statistics (ST 311)* *Fall 2021*

Awards

Glaxo Smith Klein Fellowship *2025*
Mu Sigma Rho (Statistics Honor Society) *2021*

Community Engagement

Volunteer, GenX Cohort Study, two data collection events. *2023-2024*
Interviewee, Cary Academy: Per- and polyfluoroalkyl substances (PFAS) exposure in North Carolina. *2025*

Refereeing

Philosophical Transactions of the Royal Society B
Journal of Applied Statistics: Environmental Statistics and Data Science
Annals of Applied Statistics
Statistical Methods in Medical Research